

EDUSENSE AI

Explainable Student Learning Outcome Prediction from Tutoring Conversations

AI Lab Project Proposal

Team Members

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1. Problem Statement

Tutoring conversations contain valuable information about a student's learning process. Student responses, reasoning, participation, tutor questioning, dialogue patterns, and the learning objective may all provide signals about the student's level of understanding.

However, identifying such signals manually across a large number of tutoring sessions is difficult and time-consuming. A system capable of automatically analyzing these conversations could provide useful support for educational assessment and learning analytics.

This project proposes to investigate whether machine learning and natural language processing can be used to analyze tutor–student conversations and predict whether a student is likely to answer the next question correctly.

2. Motivation

Most outcome-based educational analysis focuses on whether a student answered a question correctly or incorrectly. Such an outcome does not, however, capture the interaction that occurred before the answer.

A tutoring conversation contains both linguistic and behavioral information. For example, a student's response patterns, participation level, response length, tutor questioning behavior, and dialogue structure may provide additional information about learning progress.

The proposed project will therefore investigate whether combining conversation-level information with structured interaction signals can provide useful predictions of student learning outcomes.

3. Dataset

The project will use a tutoring-conversation dataset containing approximately **35,000 response-level samples** covering around **400 learning objectives**.

The dataset contains conversational and structured information, including:

- Learning objectives and tutoring transcripts
- Student and tutor utterances
- Session duration and conversation length
- Student participation and response characteristics
- Tutor questioning behavior
- Speaker-switching and dialogue characteristics
- Numerical and categorical session-level features

The target variable will represent whether the student's relevant answer was correct or incorrect. The prediction system will aim to produce a probability of a correct outcome rather than only a binary class.

4. Proposed Methodology

The project will follow an experimental machine-learning workflow rather than assuming a single model or representation in advance.

First, the dataset will be explored and preprocessed to understand the distribution of the target variable, conversational characteristics, and available structured features.

Next, different representation and modeling strategies will be investigated. A traditional text-based machine-learning approach may be used to establish a baseline. Transformer-based language models will also be considered for capturing contextual information from tutoring conversations.

The study will then investigate whether structured interaction features can be combined with textual representations to improve prediction.

The general experimental framework will be:

Data → Preprocessing → Representation → Modeling → Evaluation

Candidate approaches will be selected based on validation performance, computational feasibility, and their ability to capture the characteristics of the tutoring conversations.

5. Evaluation Plan

The models will be evaluated using metrics appropriate for probabilistic binary classification.

The primary evaluation metric will be **LogLoss**, since the proposed system will generate probabilities of correct student outcomes.

Additional metrics will include:

- AUROC
- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

The evaluation will also consider model calibration and prediction confidence.

Error analysis will investigate false positives, false negatives, high-confidence errors, difficult samples, and performance across different learning objectives.

6. Proposed System

The final system, **EduSense**, will provide an interactive interface for analyzing tutoring conversations.

The proposed architecture will contain three main layers:

Frontend → Backend/API → Machine Learning Pipeline

The frontend will provide interfaces for entering or analyzing tutoring conversations and viewing prediction results and analytics.

The backend will manage data processing, model inference, and communication between the user interface and the machine-learning pipeline.

The system will present:

- Predicted probability of a correct outcome
- Prediction confidence
- Conversation-level analysis
- Learning-objective-level analytics
- Error analysis
- Explainability-oriented insights

7. Expected Outcomes and Deliverables

The project is expected to produce:

- An evaluated machine-learning model for student outcome prediction
- A reproducible data preprocessing and feature-engineering pipeline
- Comparative experimental results across candidate approaches
- Error and confidence analysis
- An explainability component
- A FastAPI-based inference backend
- An interactive EduSense web dashboard
- A final technical report documenting methodology, experiments, findings, limitations, and future work

8. Timeline

Phase	Planned Activities
Phase 1	Dataset exploration, preprocessing, and baseline development
Phase 2	Representation and model investigation, followed by feature integration
Phase 3	Evaluation, error analysis, confidence analysis, and explainability
Phase 4	Backend API and machine-learning inference pipeline
Phase 5	Frontend dashboard, visualization, and system integration
Phase 6	Testing, documentation, final report, and presentation

The project will select the final modeling approach based on experimental evidence, validation performance, interpretability, and computational feasibility.